

HEMPHILL COUNTY AP, TX, USA

WMO: 722637

Lat: 35.900N

Lon: 100.400W

Elev: 730

StdP: 92.85

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 93046

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.3	-8.6	-18.9	0.8	-2.6	-16.3	1.0	1.5	12.8	10.4	11.7	11.6	2.6	20	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	38.0	20.5	37.0	20.4	35.6	20.7	24.1	32.2	23.5	31.6	22.9	30.9	6.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	18.4	27.0	21.4	17.5	26.4	20.8	16.9	26.1	76.7	32.2	74.0	31.8	71.7	31.1	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	11.0	9.6	-16.1	40.6	4.2	2.3	-19.2	42.2	-21.7	43.6	-24.0	44.9	-27.1	46.5
			-16.6	25.1	4.1	1.4	-19.6	26.1	-22.0	26.9	-24.3	27.7	-27.3	28.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	3.5	4.6	10.9	15.0	19.9	25.6	27.4	26.3	22.5	15.5	9.1	4.0
	DBStd	9.82	5.54	6.17	5.53	5.19	4.85	3.30	2.94	3.12	3.97	5.08	5.25	5.13
	HDD10.0	733	211	168	56	15	1	0	0	0	0	12	77	194
	HDD18.3	2093	458	384	237	125	41	0	0	0	10	116	278	444
	CDD10.0	2714	11	18	83	164	309	468	539	506	375	183	51	8
	CDD18.3	1034	0	1	7	24	90	218	281	248	135	28	2	0
	CDH23.3	13170	6	32	141	420	1159	2701	3683	2982	1540	431	73	3
	CDH26.7	6789	0	9	40	164	533	1419	2113	1636	707	151	17	0
Wind	WSAvg	4.0	3.3	3.8	4.4	5.1	4.5	4.7	3.8	3.5	3.8	3.7	3.7	3.5
Precipitation	PrecAvg	616	20	19	39	56	89	89	66	74	52	57	28	26
	PrecMax	902	50	59	124	276	198	231	165	159	127	205	106	100
	PrecMin	452	0	0	0	0	3	4	16	5	0	3	0	3
	PrecStd	102	15	17	34	56	41	52	36	33	33	52	29	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.9	27.2	30.6	34.2	37.1	40.2	40.9	39.9	37.1	33.1	28.9	22.7
		MCWB	10.0	11.0	13.6	13.9	16.7	18.7	19.8	21.2	19.5	16.9	12.9	9.7
	2%	DB	20.4	22.9	27.2	31.3	34.8	37.2	38.1	37.6	35.1	30.8	25.5	19.2
		MCWB	8.8	10.2	13.7	13.9	17.5	19.6	20.7	21.0	20.2	17.4	12.4	9.0
	5%	DB	17.0	19.2	24.0	27.9	32.4	35.2	37.1	36.2	32.9	27.8	22.2	16.2
		MCWB	7.3	9.1	12.1	14.4	17.5	20.7	21.0	21.3	20.2	16.5	11.8	7.4
	10%	DB	13.7	15.6	21.4	25.1	29.8	33.4	35.4	34.5	31.1	25.4	19.1	13.2
		MCWB	5.9	7.3	11.3	13.9	17.7	21.1	21.4	21.5	19.7	15.7	11.0	6.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	13.9	18.0	19.6	23.0	24.8	25.2	24.7	23.7	22.1	17.3	13.3
		MCDB	20.7	21.4	23.7	26.0	29.3	32.5	32.9	32.3	30.7	27.0	20.1	16.4
	2%	WB	9.7	11.2	16.6	18.2	21.7	23.9	24.2	23.8	22.7	20.2	14.6	10.4
		MCDB	18.0	19.9	21.8	23.5	27.5	31.8	32.4	32.0	29.7	25.5	19.6	17.1
	5%	WB	8.0	9.8	15.0	16.8	20.6	23.1	23.5	23.0	21.8	19.0	13.1	8.6
		MCDB	16.0	18.8	20.2	22.3	26.9	31.1	31.8	31.6	28.6	23.9	19.8	15.3
	10%	WB	6.4	8.0	12.8	15.4	19.6	22.4	22.8	22.5	21.1	17.4	11.6	6.8
		MCDB	13.0	15.3	19.6	22.0	26.3	30.1	30.9	30.5	27.6	22.8	18.6	13.2
Mean Daily Temperature Range	5% DB	MDBR	15.3	14.7	15.5	15.0	14.2	13.4	13.6	13.6	14.0	15.4	15.9	14.5
		MCDBR	20.5	21.5	19.3	19.6	17.7	15.7	15.7	16.0	15.9	19.1	20.7	19.5
		MCWBR	10.8	10.9	8.8	8.1	6.5	4.9	3.9	4.2	5.1	8.1	10.0	10.9
	5% WB	MCDBR	19.0	19.8	15.1	14.1	14.2	13.3	13.4	13.6	12.8	13.9	16.4	17.1
		MCWBR	10.5	11.0	7.9	7.5	6.6	5.4	4.7	4.5	5.0	7.3	9.2	10.1
Clear-Sky Solar Irradiance	taub	0.267	0.274	0.309	0.339	0.356	0.392	0.395	0.384	0.351	0.318	0.286	0.269	
	taud	2.565	2.555	2.464	2.402	2.424	2.365	2.391	2.383	2.456	2.526	2.552	2.567	
	Ebn at Noon	950	979	963	944	926	888	883	888	906	913	917	926	
	Edh at Noon	76	86	104	116	116	123	119	118	104	88	76	71	
All-Sky Solar Radiation	RadAvg	2.97	3.73	4.93	6.09	6.77	7.31	7.16	6.39	5.44	4.06	3.15	2.57	
	RadStd	0.27	0.40	0.36	0.40	0.58	0.51	0.33	0.34	0.27	0.51	0.33	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+109	N/A

Nomenclature: See separate page